Exp no: 14
11/10/26

Implement packet sniffing using Raw sockets

Aim: Write a code to using raw sockets to implement packet sniffing.

Code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        # Determine the protocol
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"

        # Print packet details
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)

# Capture packets on the default network interface
sniff(iface='Wifi', prn=packet_callback, filter="ip", store=0)
```

Output:

Protocol : TCP
Soure IP : 57.155.141.117
Destination IP : 192.168.1.6

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Destination IP : 57.155.141.117

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Protocol : TCP
Soure IP : 57.155.141.117
Destination IP : 192.168.1.6

Protocol : UDP
Soure IP : 192.168.1.6
Destination IP : 192.168.1.1

Protocol : UDP
Soure IP : 192.168.1.6
Destination IP : 192.168.1.1

Protocol : UDP
Sourc IP : 192.168.1.6
Destination IP : 192.168.1.1

Protocol : UDP
Sourc IP : 192.168.1.61
Destination IP : 192.168.1.6

Result : Successfully Implemented packet sniffing using raw sockets.